Swedish Competence Centre for Satellite-Enabled Social Science Analytics

Rymdstyrelsen
Swedish National Space Agency

Beginner Earth Observation Analysis

Hand-on Workshop: From Flood Mapping to Critical Infrastructure Accessibility

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SESAC, Project Assistant

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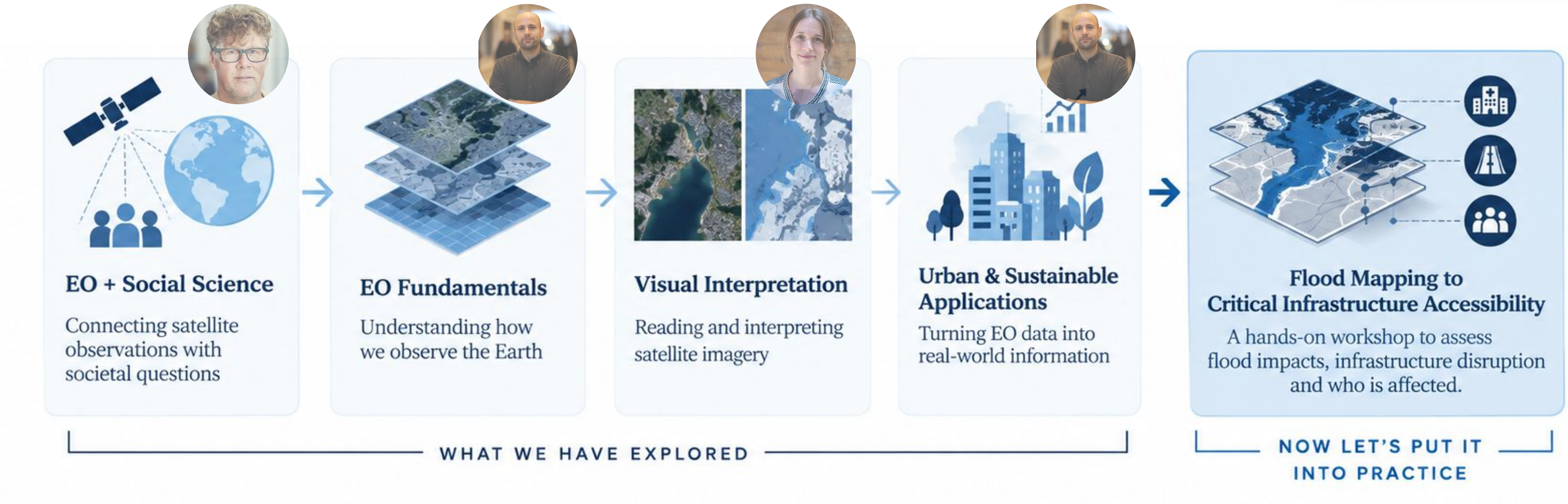
Scan to access the materials

For EO fundamentals to real world applications

Building on what we have already learned



Let’s start with you

Mentimeter activity #01

When you think about **flood impacts**, what comes to mind first?

Enter 1–3 words or short phrases.

There are no right or wrong answers.

Mentimeter

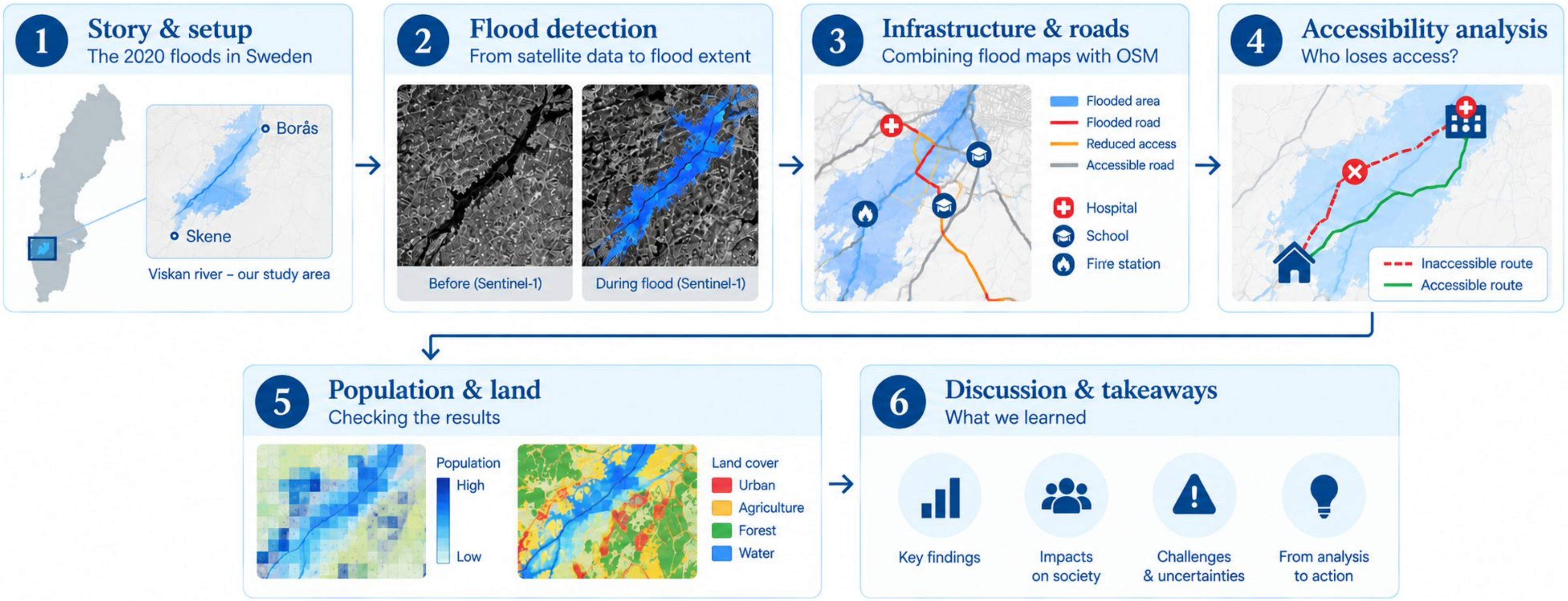
Go to **menti.com**

Enter the code

3883 4156

Workshop overview

From satellite signal to societal applications



The 2020 floods

A large-scale flood event in southern Sweden

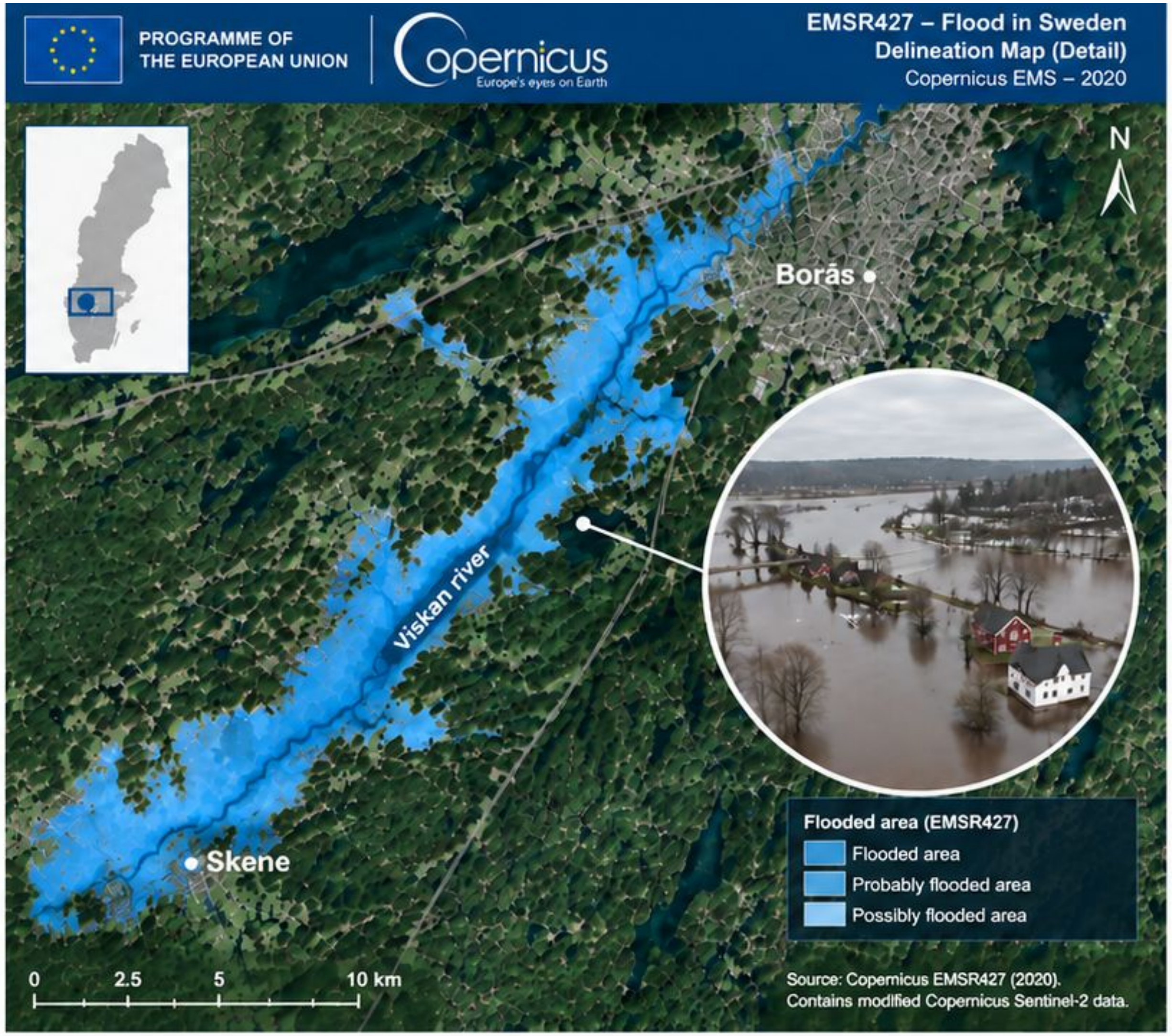
18 February 2020
Copernicus Emergency Management Service activated (EMSR427)

12 areas · ~ 41,200 km²
mapped across southern Sweden

Sjuhärad / Viskan river
Borås → Skene
Our focus today

The Viskan river was **one of the two rivers** that EFAS flagged with the **highest flood risk** in this event.

Widespread flooding affected large parts of southern Sweden in February 2020, impacting communities, infrastructure and daily life.



Data sources

Real world data for real world impacts

DATA SOURCE	PROVIDER	ACCOUNT NEEDED?
Sentinel-1 GRD SAR (VH)	Copernicus Data Space Ecosystem (Sentinel Hub)	Yes
Roads + hospitals/fire/police/schools	OpenStreetMap (osmnx)	No
Population grid	SCB (Statistics Sweden) 1 km grid — GitHub repo	No
Land cover raster	NATUR VÅRDS VERKET NMD2018 (Naturvårdsverket) — GitHub repo	No

The population and land-cover rasters are already pre-clipped for our study area and hosted on GitHub, so no account is needed and no live download is required.

Creating a Sentinel Hub Account

One quick setup before we start working with

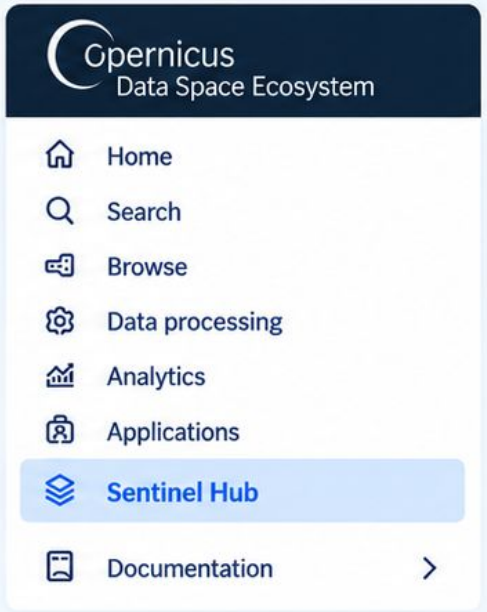
01 Create an account

Go to the Copernicus Data Space Ecosystem and sign up (or log in if you already have an account).

https://dataspace.copernicus.eu

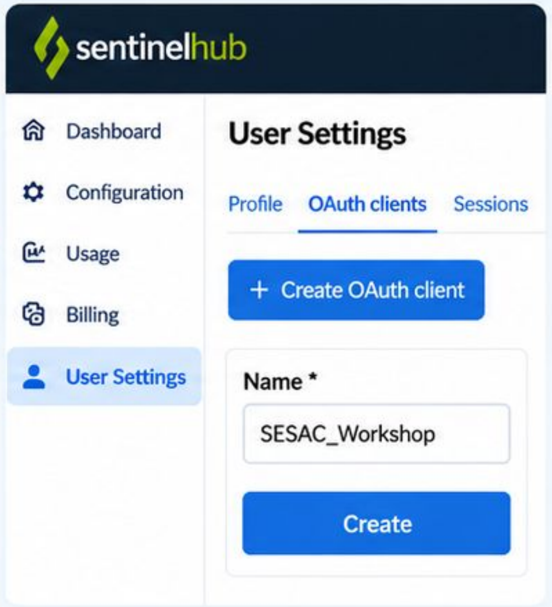
02 Open Sentinel Hub

After logging in, go to the Sentinel Hub Dashboard.



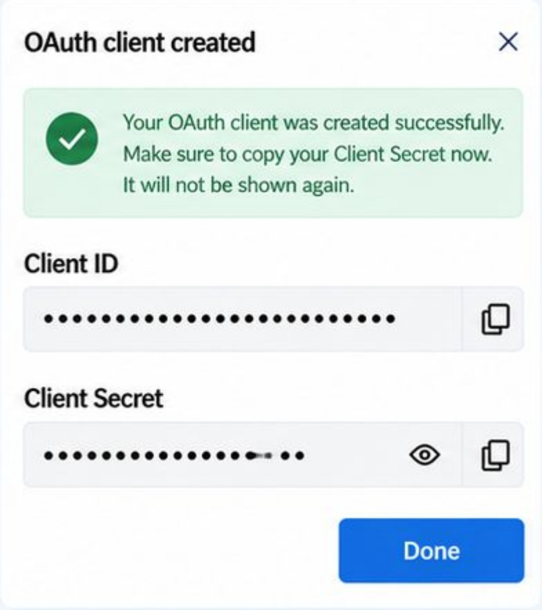
03 Create an OAuth client

In your Sentinel Hub account, go to User Settings → **OAuth clients** and click Create.



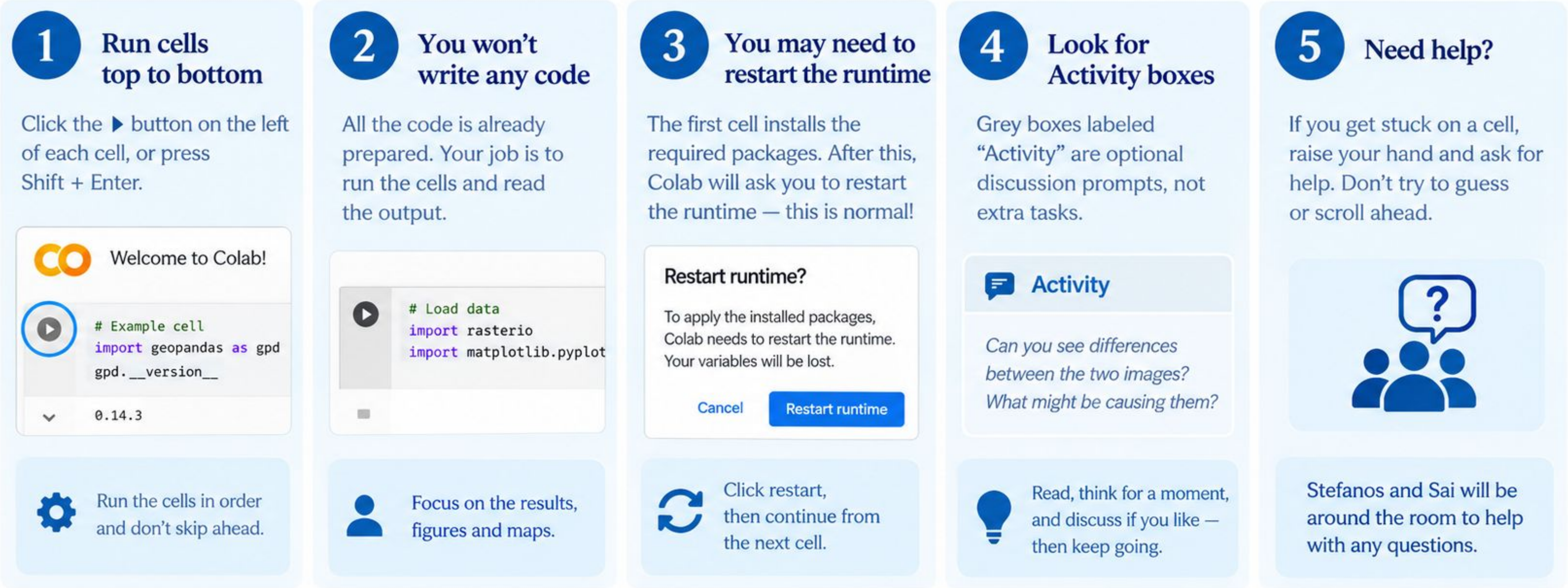
04 Save your credentials

After creating the client, copy your Client ID and Client Secret. You will use these in the notebook.



Using Google Colab Notebook

A few quick tips to get the most out of the hands-on session



How we detect flooding from Sentinel-1

A simple and robust rule

1. Low backscatter

The pixel backscatter is below –17 dB.

$\sigma^0 < -17$ dB

AND

2. Significant drop

The backscatter decreased by at least 3 dB compared to the baseline (pre-event).

$\Delta\sigma^0 \leq -3$ dB

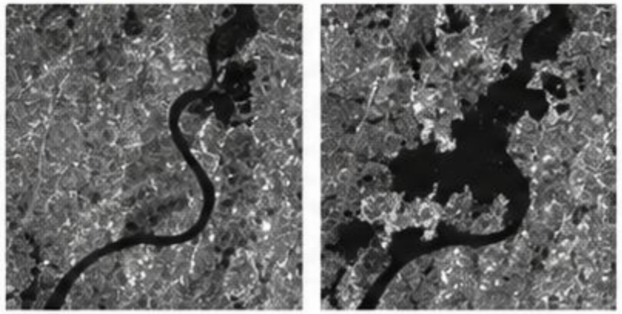
→

Pixel classified as flooded

Flooded (1)

Not flooded (0)

Example (Sentinel-1 backscatter)



Before (baseline) — During flood

Detected flood (binary mask)

Why both conditions?

Using both an absolute threshold and a significant drop ensures that we detect new flooding, rather than permanent water bodies (e.g. lakes or large rivers) that always appear as dark in SAR images.

What this method can (and cannot) tell us

Strengths and limitations of our flood detection formula

What it can tell us

Detects where water is present
Sentinel-1 SAR is sensitive to open water surfaces, allowing us to map inundated areas.

Identifies newly flooded areas
By comparing pre- and post-event images, we detect areas that became water-covered.

Provides consistent, large-scale coverage
Works under cloudy conditions and over large areas.

Useful for understanding impacts on the road network
Helps us identify which road segments are affected and may become unusable.

What it cannot tell us

It does not provide water depth
SAR measures surface reflectance, not the height or depth of the water.

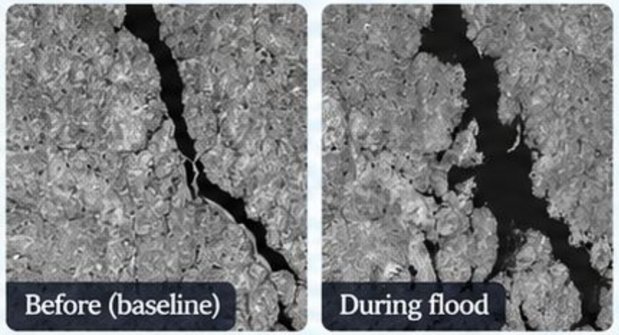
We do not use terrain information (e.g. slope/DEM filtering)
Some detections may occur on flat, low-lying areas that are normally wet (e.g. permanent water bodies).

A road is treated as either completely unavailable or fully available
We do not model partial obstruction (e.g. shallow or muddy water).
The classification is binary: flooded or not flooded.

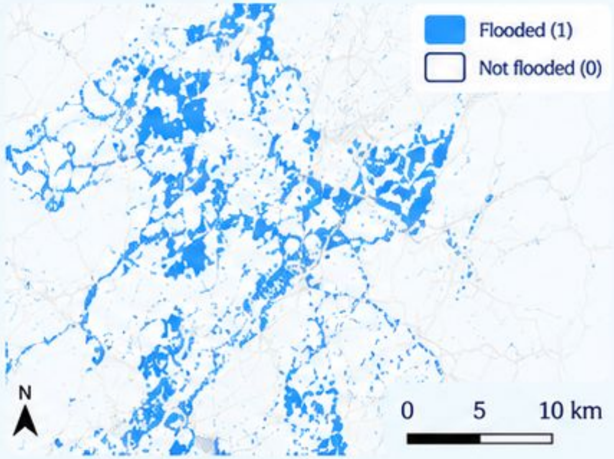
Some static water bodies may still appear as flooded
Although we use a change detection approach, very small or temporal changes in backscatter can occasionally lead to false detections.

In this workshop, we use a simplified, binary representation of flooding, based on Sentinel-1 observations.

Sentinel-1 backscatter (VV)

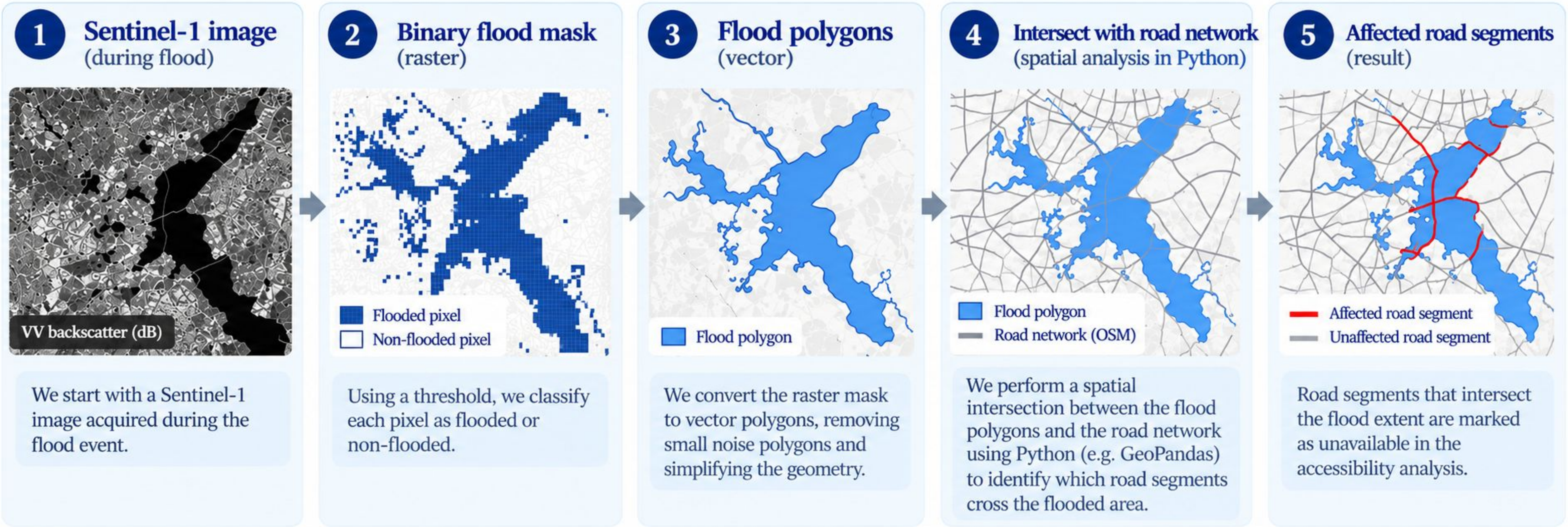


Detected flood extent



From pixels to network disruption

Turning an EO observation into something we can analyse in Python

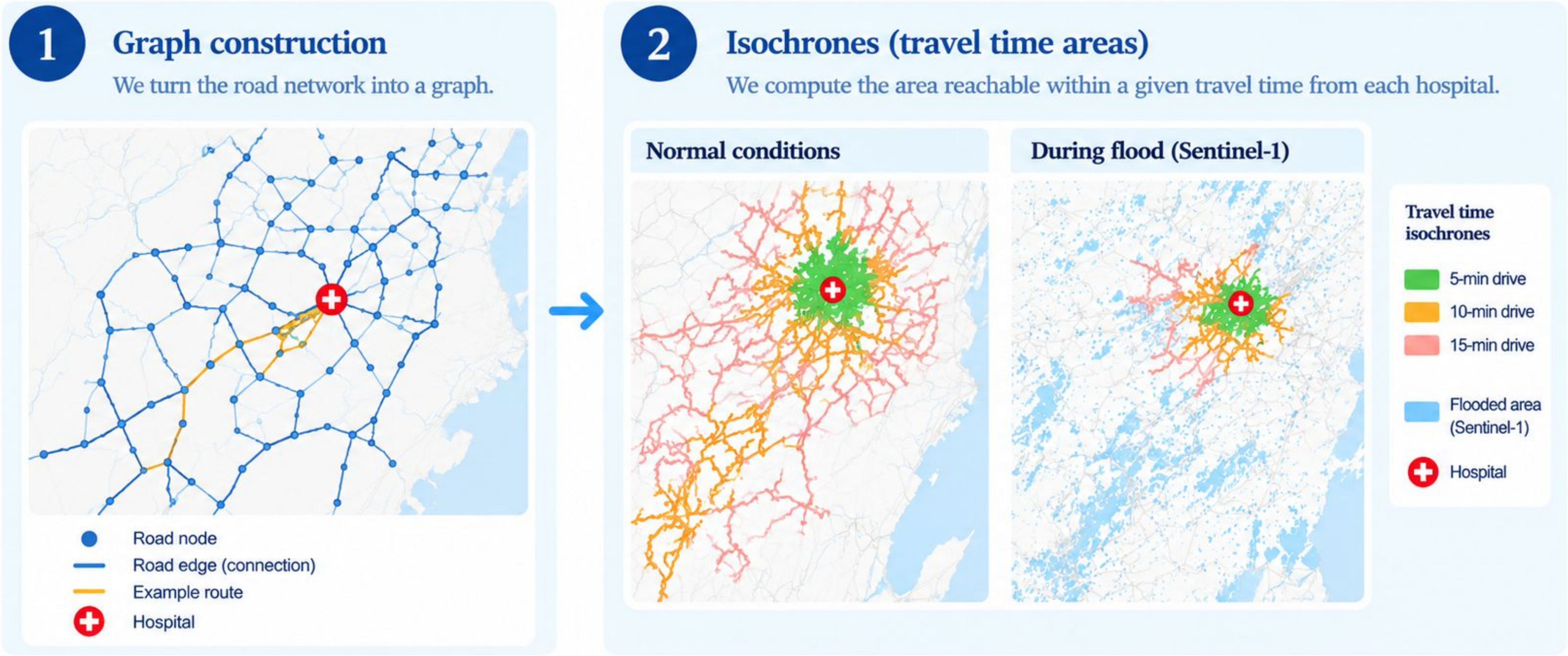


Key takeaway

The satellite result is initially a raster of flooded pixels, while roads are vector geometries.
By converting the flood mask to polygons, we can combine the two in Python and identify which road segments are affected.

From road to accessibility

Graph construction and isochrones



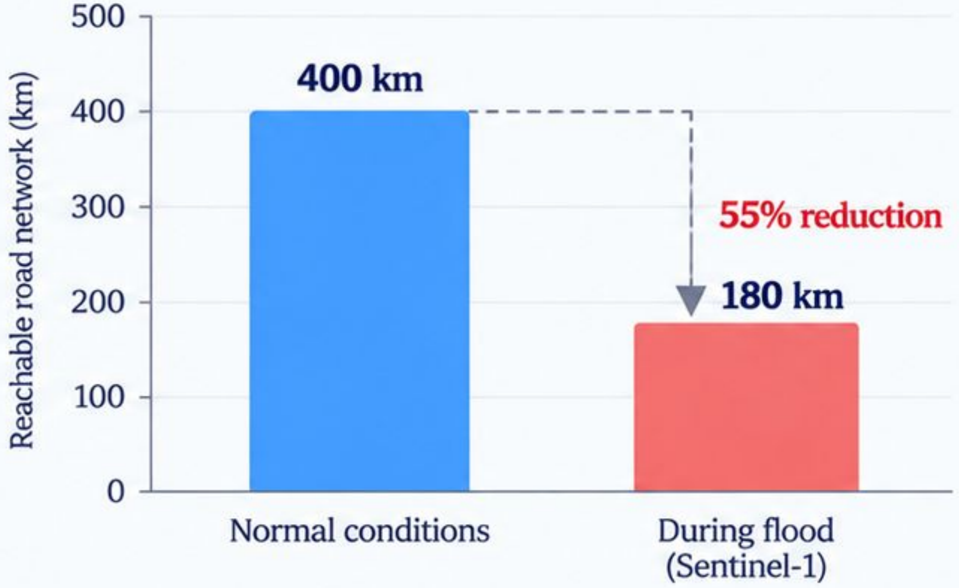
Let’s hear from you

Mentimeter activity #02

Quantifying the impact on accessibility

Accessibility reduction formula

Example: 10-min accessibility to Hospital



Formula

$$\text{Accessibility reduction (\%)} = \frac{L_{\text{normal}} - L_{\text{flood}}}{L_{\text{normal}}} \times 100$$

L_{normal} = total length of the reachable road network within the selected travel time under normal conditions (km)

L_{flood} = total length of the reachable road network within the selected travel time during flood conditions (km)

How to interpret the results

0%
No reduction
The reachable network is the same as in normal conditions.

~100%
Almost no access
Very little (or none) of the network remains reachable.

Key takeaway

The accessibility reduction tells us how much the reachable road network is affected by flooding. A higher percentage means a stronger impact on access to the facility, even if the facility itself is not directly flooded.

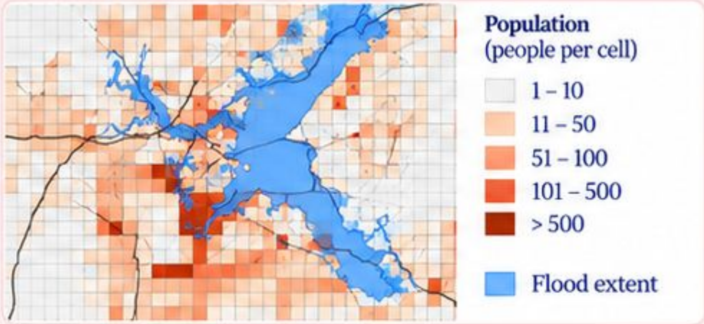
From accessibility to exposure

Adding the human and environmental layers

Flood extent and disrupted access are important, but the key question is:
Who and what is affected?

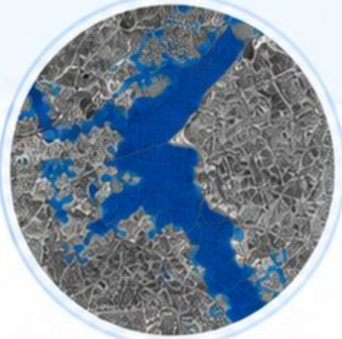
Population
Who is affected?

- **Direct exposure**
  How many people live within the flood extent?
- **Accessibility exposure**
  How many people lose previously available access to a hospital (e.g. within 10 minutes)?
- **Different travel times**
  Compare 5, 10 and 15 minutes.



This helps us move from kilometres of affected roads to people who may be impacted.

Flood extent
(from Sentinel-1)



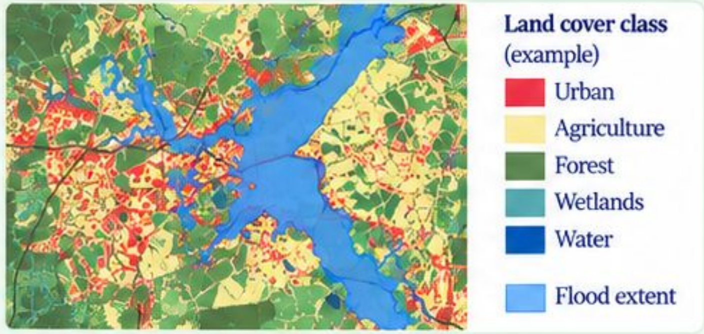
So what?

Towards a more complete picture of flood impacts

People	Services	Environment
Who is affected?	What is disrupted?	What areas are affected?

Land cover
What environment is affected?

- **Which land cover classes are flooded?**
  e.g. urban areas, agricultural land, forests, wetlands, water bodies.
- **Which classes are disproportionately affected?**
  Understand the environmental context of the flooding.
- **Where might SAR perform less well?**
  Some surfaces (e.g. vegetation, urban areas) can lead to underestimation of flooding.



Land cover provides environmental context and also helps assess the limitations of the Sentinel-1 detection.

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Thank you! ☺

SESAC · AUTUMN 2026

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